

Expression of Interest for Flagship Scaling Solution Track

Section 0 – Title and contact Information

- **Title of the proposed solution track:** Scaling bundled digital financial services for smallholder farmers to foster agricultural technology adoption
- **Title of the AoW2 Flagship that the solution track aligns with:**
 - 1.1 (Scaling Digital Climate Services for Farmers and Enterprises)
 - 2.1 (Scaling Seed and Fertilizer Soil Fertility Management Systems for Resilient, Diverse, and Nutrition-Smart Farming)
- **Lead Scientist(s) / Contact Person(s) / Center(s):** Berber Kramer, IFPRI; Julian Ramirez-Villegas, Alliance of Bioversity International and CIAT
- **Emails of the Lead Scientist(s) / Contact Person(s) / Center(s):** b.kramer@cgiar.org, j.r.villegas@cgiar.org
- **Key CGIAR Center Partners (in the case of multi-Center Eols):** IFPRI, ABC
- **Technical link to other Science Programs:** Breeding for Tomorrow, Gender Accelerator, Climate Action

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

- **The problem to be solved:** Limited access to reliable and affordable financial services is a major barrier to smallholder farmers' use of agricultural innovations that reduce climate risk, including quality seeds of improved climate-adapted varieties and fertilizers. Documented land ownership can be crucial in obtaining financial services, because land titles are often required as collateral to receive loans from financial institutions and to determine a farmer's eligibility for government-sponsored agricultural credit and insurance. This is a challenge, especially for our proposed target group of smallholder farmers with **relatively limited land ownership**, including women and youth farmers, sharecroppers, and farmers of lower socioeconomic status. We propose scaling bundled digital financial services co-developed and tested by CGIAR and two social enterprises. These bundled digital financial services have been developed and validated to directly address this challenge around limited land ownership in **Kenya** and **India**.
- **Your scaling ambition:** The proposed Scaling Solution track aims to provide improved financial services to 100,000 smallholder farmers in Kenya and India combined (KPI #1). At least 50 percent of beneficiaries will be women farmers and 25 percent will be youth (under 35 years old). These groups are oftentimes marginalized and underrepresented in existing agricultural credit and insurance schemes (KPI #2), and formative studies in [Kenya](#) and [India](#) have demonstrated that these products are especially in-demand and impactful for them. At least \$ 25,000,000 will be disbursed and insured (KPI #3). In this way, we expect the solution to support livelihoods for close to 500,000 people from 100,000 farm households across Kenya and India. Based on prior randomized impact evaluations in [Kenya](#) and [India](#), we expect to achieve the following potential impacts: increased adoption of climate-smart technologies and practices on 100,000 hectares; increased hiring of labor, enhancing job prospects in targeted locations; improved

agricultural incomes, translating to healthier diets; and enhanced women's empowerment and mental health.

- Innovation & Value Proposition:** The innovation is a bundled [credit-insurance solution](#), co-developed by IFPRI and scaling partners as part of the CGIAR Research Programs on Policies, Institutions and Markets (PIM) and on Climate Change, Agriculture and Food Security (CCAFS), and later **Ukama Ustawi**, **S4I**, and the **Gender Accelerator**. The solution determines a farmer's creditworthiness by combining plot-level climate and soil health data with remote sensing and machine learning, developed using CGIAR data for ground truthing. This is linked with picture-based insurance, pioneered by IFPRI to compensate farmers for visible crop damage from natural causes that could hamper loan repayment, thereby protecting financial institutions from production risks that could lead to defaults. IFPRI-led randomized trials with approximately 5,000 farmers in [two states in India](#), and seven counties in [Kenya](#), validated the solution, proving its maturity and suitability for scaling. These trials demonstrated strong impacts on end-users' financial inclusion, agricultural incomes, empowerment, and mental health, particularly among women.
- Customers & Alignment:** Institutional customers include social investors, microfinance institutions (MFIs), banks, and insurers to finance and underwrite the product, and aggregators such as MFIs (Kenya) and farm producer organizations (FPOs, India). Farmers are our end users. By increasing farmers' access to finance and de-risking agricultural production, the solution can encourage climate-smart investments in seeds, fertilizers, irrigation, and other inputs. As such, this digital solution links to **Flagship 1.1** (and the Grand Challenge of closing the climate adaptation gap through scalable climate services and smart water management) and **Flagship 2.1** (and the Grand Challenge of delivering inputs and services at scale through inclusive market systems).
- Evidence of Impact:** Randomized trials have demonstrated scaling readiness and impacts of these financial services within our target groups. In [Odisha](#), access to digital credit bundled with picture-based insurance (via Dvara E-Registry's KhetScore) increased borrowing from formal sources, insurance take-up and renewal, with households reporting fewer repayment difficulties. This increased profitability in the monsoon season and irrigation use—and thus production—in the dry season. The bundle improved women's empowerment and mental health, showing that expanding financial inclusion can enhance both economic and wellbeing outcomes. In [Kenya](#), picture-based insurance dramatically boosted demand—particularly among women and in drought-prone ASAL counties—while also improving perceptions of trust and product quality. This encouraged climate-smart agricultural investments in drought-prone areas. These impacts have translated into institutional uptake beyond pilots: [ACRE Africa](#) has rolled out PBI products [commercially](#), while Dvara is currently providing KhetScore to 23,000 farmers [partnering with financial institutions](#).

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

- **Scaling Pathway & Sustainability.** Scaling will be conducted through a private sector business model in both India and Kenya. In India, Dvara E-Registry aggregates farmers in FPOs and employs agents to photograph farmers' land at multiple stages of the growing season. This documentation, combined with data obtained through remote sensing, determines creditworthiness and insurance payouts. Dvara is already lending to 15,000 farmers, and plans to expand to another 35,000 farmers in the next three years. Revenues outweigh costs: operational costs are 7 percent of loan value, and 5 percent is set aside as risk capital, whilst revenues generated from interest and FPO services are 10 and 5 percent of the loan value, respectively. Farmers pay their full insurance premium of 5 to 10 percent (depending on the crop) as part of loan repayment.

ACRE Africa will scale the solution through its TARA platform, which combines remote sensing, climate data, and picture-based crop monitoring to provide insights on production risks for crops in targeted locations. ACRE partners with financial institutions that use this platform to inform their lending decisions and to provide insurance against production risk to further de-risk agricultural loans. ACRE earns 20 percent commission on premiums collected, which in turn are around 10 percent of the value of loans issued through TARA. Revenues are sufficient to cover critical operational expenses such as agent commissions, capacity building, and service delivery amenities. Long-term sustainability will require risk pooling and recovery of fixed costs associated with TARA hosting and maintenance, which critically depends on achieving scale.

- **Partnerships & Leverage:** CGIAR researchers championing and scaling the bundled digital financial services include Berber Kramer (IFPRI, Breeding for Tomorrow), Shalini Roy (IFPRI, Gender Accelerator), Julian Ramirez, and Harold Achicanoy (Alliance, Climate Action). Findings from research on the feasibility of picture-based crop insurance, co-led by Dr. Kramer, led to requests for collaboration from Dvara and ACRE to co-design and test solutions linking crop insurance with digital credit (India) and seeds of improved varieties (Kenya). This resulted in two large-scale IFPRI-led impact evaluations of the digital financial services. Dr. Roy joined the team to evaluate impacts on intimate partner violence and will champion a gender-responsive and socially inclusive scaling strategy. Using past data from the randomized evaluations in India and Kenya, Julian Ramirez and Harold Achicanoy have refined Dvara's and ACRE's computer vision models to strengthen automated damage detection from smartphone images. Use of these models will further increase the accuracy and speed of crop monitoring and insurance claims settlement.

Main **scaling partners** are **Dvara E-Registry**, an [award-winning](#) agri-fintech company using digital technology to provide financial and agricultural services to small and marginal farmers in India, and **ACRE Africa**, a social enterprise that designs and implements risk management solutions for smallholder farmers across Africa. Dvara is scaling its credit-insurance solution over the next three years from an existing client base of 15,000 farmers to an additional 35,000 targeted farmers, leveraging partnerships with financial institutions such as HDFC Ergo General Insurance and Avanti Finance. A non-banking financial company license will further increase reach, profitability, and tailoring of financial services. ACRE has pioneered innovative approaches, such as picture-based insurance and mobile money platforms, to make insurance more accessible. It aims to reach 50,000 farmers with an interlinked credit-insurance solution by deploying TARA with MFIs such as Juhudi and banks such as NCBA, with underwriting by APA and Britam. TARA is supported by ACRE's shareholders Zep-Re and Crédit Agricole.

- **Risks & Bottlenecks:** Uncertainty around climate change and changes in the suitability of cultivating certain crops discourages financial institutions from lending to smallholder farmers. As a mitigation strategy to address this bottleneck, Dvara and ACRE are **integrating climate data** into their risk analytics, providing financial institutions with high-resolution data of crop suitability in targeted locations. This is expected to de-risk lending to the agricultural sector, crowding in finance and allowing Dvara and ACRE to expand the scale of their financial services.

Although both Kenya and India have large-scale insurance programs, these programs do not reach small and marginal farmers—land title requirements, limited reach and coverage, and delayed claims settlement, have eroded farmer trust and leave them exposed to risk. Our digital financial services provide complete insurance coverage (using pictures for claims settlement) and use digital technology to verify cultivation, issue loans and insurance, and crowd-in farmer trust. Close **policy engagement** will ensure that our private sector scaling efforts are complementary and maximize future policy uptake.

The perceived risk of lender default also presents a major bottleneck. Our solution embeds a combination of **first loss default guarantees (FLDGs)** and the requirement of **high-quality, reliable insurance coverage for any farmer receiving credit**. FLDGs will incentivize institutional lenders and financial institutions to lend to this new market segment. Bringing insurance premiums into loan repayment is proven to help address a major barrier to insurance uptake. The high-quality coverage offered by picture-based insurance will ensure high demand and loyalty to the financial service provider, with the potential to enhance loan repayment.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

- **Readiness & Responsible Scaling.** The ISPR score of this innovation is 9, with the innovation validated for its ability to achieve impact under uncontrolled conditions. As farmers not involved in developing the innovation are commonly using it, our use score is also 9. Dvara has already reached 15,000 farmers with its KhetScore solution, and ACRE has piloted the TARA platform with NCBA and SBM banks, which both now have requested to use the tool to scale agri-finance. To ensure the social inclusivity of the scaling initiative, this initiative targets small and marginal farmers including at least 50% women and 25% youth farmers. Our evidence shows that these target groups value and benefit from the solution most. Environmental sustainability will be safeguarded through recommendations around crop choice and input use, ensuring that farmers cultivate crops that are sustainable in the presence of climate change, without using inputs that may harm the environment.
- **Achievable target, Resources & Activities** – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships. Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

Both Dvara and ACRE plan to provide bundled digital financial services to at least 50,000 farmers in the next three years, leveraging existing operations and the following activities:

1. Engagement of policymakers, social investors, MFIs, and (re-)insurers on impacts and scalability of the solution, to further increase institutional demand for the solution and facilitate long-term sustainability.
2. Further integration of climate risk analysis in the solution. Both Dvara and ACRE have climate risk analysis embedded into their risk scoring platforms. S4I support would facilitate further refining these analytics.
3. Smart subsidies and First Loss Default Guarantees (FLDGs), phased out over time, to increase trust between financial institutions and farmers by encouraging lending to farmers and creating positive product and repayment experiences.
4. Training financial institutions and agents will be crucial to ensure understanding of how the bundled solution de-risks agricultural lending to expand financing for smallholder farmers.
5. Refining computer vision models for faster and more accurate loss assessment. Dvara and ACRE have computer vision models in place for picture-based damage detection; additional training data can further enhance the solution.
6. Bundling with customized climate-informed advisories, tailored to reduce risks for our target groups, especially women. This will include recommendations on the use of high-quality, climate-smart, and environmentally sustainable inputs.

Resources required:

1. Operational funding for partners for training of field staff and financial institutions, smart subsidies, and FLDGs, among other purposes
2. CGIAR staff – lead scientists and program coordinator to oversee responsive scaling
3. Convening: Travel, conferences, and meetings for stakeholder engagement
4. Monitoring, evaluation, and learning, such as strengthening the algorithms used for claims settlement using computer vision models

• **Timeline, Milestones & Monitoring**

2026: We will have formalized the two partnerships (contracts with ACRE and Dvara fully executed) and trained partnering financial institutions. Both Dvara and ACRE will have provided the bundled digital financial service to 12,500 farmers, reaching a total of 25,000 farmers, with loan disbursements of at least \$ 6,000,000. Doing so, we leverage established partnerships with financial institutions (MFIs, social investors, and insurers).

2027: An additional 30,000 smallholder farmers will access the bundled solution. This will be associated with an additional \$ 8,000,000 of loans disbursed and protected. Monitoring, evaluation and learning will be used to generate impact stories to inform donors of the benefits achieved through Scaling for Impact.

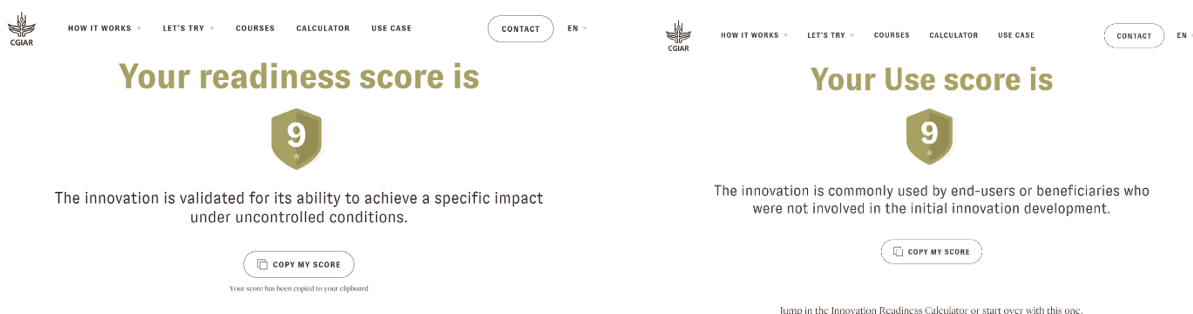
2028: Both Dvara and ACRE will have reached an additional 22,500 farmers, disbursing and protecting another \$11,000,000 worth of loans, with demonstrated impact. Two new

partnerships with financial institutions will be established, informed by evidence of impact and cost effectiveness, to facilitate further expansion beyond this initiative's lifespan.

Monitoring & Learning: Progress will be tracked by monitoring progress towards a detailed theory of change that maps activities into outputs and outcomes, linked to the milestones and KPIs above. Quarterly reviews with partners will allow for adaptive management, while annual reports will capture lessons and inform replication.

Optional Annexes

Scaling Readiness assessment results



Media or other evidence of prior scaling work that can be accelerated

[The Impact Of Field Photography On Crop Insurance Development In Kenya | ACRE Africa](#)

[ACRE Africa provides micro-insurance products to strengthen farmer resilience | Private Finance for the SDGs \(UNDP\)](#)

[Kenyan farmers champion crop insurance for climate resilience | IDRC - International Development Research Centre](#)

[Financial Institutions - Dvara E-Registry | Integrated Solutions for Agri Value Chain](#)

[KhetScore: Impacts of Using Digital Technologies to Unlock Credit and Insurance for Marginal Farmers in Odisha](#)

[Scaling credit for India's small and marginal farmers—₹150 crore disbursed and growing](#)

[Dvara E-Registry has won the 1st Place at the International Climate Competition "Green Eurasia-2025", under the category Climate-Oriented Agriculture and Forestry.](#)

[Dvara E-Registry has been honoured with the prestigious BRICS Solutions Award in the Climate and Environmental Technologies category](#)

[Small holder farmers benefit from use of technology, financial inclusion shows survey | India News - Times of India](#)

MELIA studies, impact assessments, reports, publications

- Ceballos, F., Kramer, B., and Robles, M. (2019). "The feasibility of picture-based insurance (PBI): Smartphone pictures for affordable crop insurance." *Development Engineering* 4: 100042.
- Kramer, B., Pattnaik, S., Ward, P.S. and Xu, Y. (2025). "Bundling crop insurance with digital credit to boost agricultural investments: Findings from a randomized trial in two states in India." Paper prepared for presentation at the 2025 AAEE & WAEA Joint Annual Meeting in Denver, CO; July 27-29, 2025.
- Kramer, B., Pattnaik, S., Ward, P.S. and Xu, Y. (2024). "Impacts of an innovative credit+ insurance bundle for marginalized farmers: Evidence from a cluster randomized trial in Odisha, India." IFPRI Discussion Paper 2288. Washington, DC: International Food Policy Research Institute.
- Kramer, B.; Pattnaik, S.; and Ward, P.S. (2021). "Gender, demand for agricultural credit and digital technology: Survey evidence from Odisha." IFPRI Discussion Paper 2093. Washington, DC: International Food Policy Research Institute (IFPRI).
- Kramer, B., Waweru, C., and Malacarne, J.G. (2023). "Control over future payouts and willingness-to-pay for insurance: Experimental evidence from Kenyan farmers." Selected Paper prepared for presentation at the 2023 Agricultural & Applied Economics Association Annual Meeting, Washington DC; July 23-25, 2023.
- Kramer, B., Waweru, C., Cecchi, F., Kivuva, B., and Waithaka, L. (2024). "See It Grow: A randomized evaluation of digital innovations in crop insurance to increase insurance and fertilizer demand in Kenya." Selected Paper prepared for presentation at the 2024 Agricultural & Applied Economics Association Annual Meeting, New Orleans, LA; July 28-30, 2024.

Partner commitments (letters of support) – see next pages

Date: 19th September 2025

To Whom It May Concern,
Subject: Letter of Support for IFPRI's Impact Scaling Project Proposal

We are pleased to extend our strong support for the International Food Policy Research Institute's (IFPRI) Expression of Interest for a Scaling Solution Track entitled "providing digital financial services for smallholder farmers to foster agricultural technology adoption at scale".

Dvara E-Registry (DER) has partnered with IFPRI for over six years through projects supported by CGIAR, the International Initiative for Impact Evaluation (3ie), the Digital Credit Observatory, and the Wellspring Philanthropic Fund. This work has been instrumental in strengthening our digital solutions to expand access to agricultural credit and crop insurance for smallholder, tenant, and women farmers often excluded from formal finance. Our recent collaboration with IFPRI on KhetScore, DER's innovative credit scoring methodology, shows that digital credit scoring bundled with crop insurance can improve finance access, reduce risks, and boost farm investments. By embedding picture-based insurance in our loans, under real-world uncontrolled conditions, we demonstrated feasibility and added value of this transparent and cost-effective approach to claims settlement.

These initiatives highlight the scaling readiness and impact potential of our digital solutions to strengthen resilience and empower marginalized farmers through financial inclusion. IFPRI has brought global research expertise, ground truth data, insights on gender programming, and objective evidence of impact, while DER has contributed through its field implementation, digital platforms (including KhetScore, our innovative digital credit scoring approach, and Doordrishti, a platform through which we provide agricultural advisories), and financial sector partnerships. The proposed project is a timely opportunity to scale proven innovations.

DER remains committed to supporting this effort through field engagement, data integration, and partnerships with financial institutions and policymakers. We are confident the project will deliver lasting impact on financial inclusion, climate resilience, and agricultural productivity in rural India.

Sincerely,

Tarun Katoch

Tarun Katoch
Co-founder & Head Value Chain
Dvara E-Registry Private Limited



Providing Innovative Insurance
Solutions to Farmers Across Africa

To Whom It May Concern,

Letter of Support – Expression of Interest: Providing Digital Financial Services for Smallholder Farmers to Foster Agricultural Technology Adoption at Scale

Agriculture and Climate Risk Enterprise Limited (ACRE Africa) is pleased to express its strong support for the CGIAR-led Expression of Interest on *Providing Digital Financial Services for Smallholder Farmers to Foster Agricultural Technology Adoption at Scale*.

ACRE Africa brings in over a decade of experience in designing and delivering inclusive risk management solutions across Sub Saharan Africa. Our digital platforms, including our **Tool for Agriculture Risk Assessment & Advise (TARA)**, enable financial institutions to extend services and credit to last-mile farmers, while integrating climate data to de-risk agricultural lending. We have also co-developed and tested a picture-based insurance solution with CGIAR and other partners, demonstrating both scalability and measurable impact on farmer resilience, input adoption, and household welfare.

We recognize the alignment of this initiative with our mission to safeguard smallholder livelihoods against climate-related risks while driving sustainable market growth. We are committed to supporting the consortium through knowledge sharing and product scaling pathways with our financial and insurance partners in Kenya. Based on evidence from our past research with IFPRI, we firmly believe that TARA's impacts are strengthened by interlinking credit facilitated through the TARA platform with the affordable, high-quality crop insurance solution that we have co-developed and evaluated with CGIAR.

We strongly endorse this EoI and look forward to collaborating with CGIAR and partners to ensure that we strengthen impacts for the 50,000 farmers targeted through TARA over the next three years. This project will enable us to provide them with the financial tools they need to adopt agricultural technologies at scale, improve resilience, and achieve sustainable livelihoods.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'Ewan Wheeler'.

Ewan Wheeler
Chief Executive Officer
Agriculture and Climate Risk Enterprise Limited (ACRE Africa)



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